INTERIM FEDERAL SPECIFICATION

PADLOCK, KEY OPERATED (RESISTANT TO OPENING BY FORCE, PICK, AND BYPASS TECHNIQUES)

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of key operated, dead bolt padlocks (see 1.2.1). The padlocks provide a relatively high degree of resistance to opening by force and tamper techniques (see 6.1). The padlocks are supplied with and without chain and clevis.

1.2 Classification.

1.2.1 Types and styles. The padlocks shall be of the following types and styles, as specified (see 6.2).

Type ES - Exposed shackle.
Type SS - Shrouded shackle.
Style 1 - With chain and clevis (see 3.9).
Style 2 - Without chain and clevis.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

RR-C-271 - Chains and Attachments, Welded and Weldless, and Roller Chain.
SS-G-659 - Graphite; Lubricating, Flake.
UU-P-553 - Paper, Wrapping Tissue.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxe , Fiberboard.
PPP-B-640 - Boxes, Folding, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:47Z
Check the source to verify that this is the current version before use.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case and Sheet, Overwrap, Water-Vaporproof, Flexible.
MIL-M- 7866 - Molybdenum Disulfite, Technical, Lubrication Grade.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The padlocks furnished under this specification shall be products which have been inspected and tested, and passed the qualification tests specified in section 4 herein, and have been listed on or approved for listing on the applicable Federal Qualified Products List (QPL).

3.2 Material. Material used in the padlock's construction shall be as specified herein. Material not definitely specified shall be of good commercial quality, suitable in all respects for the purpose intended.

3.2.1 Chains. Chains shall conform to wire link or stamped link of RR-C-271.

3.3 Design and construction. The padlock's design and construction shall be as specified herein.

3.3.1 General. The padlocks shall be key operated using a single key. The padlocks shall be the dead-bolt type and shall be individually keyed with a minimum of 100,000 different, independently secure key changes. The total weight of the padlock shall not exceed 3 pounds.

3.3.2 Padlock shackle. The shackle shall have a cutaway at both heel and toe to permit mating with the locking bolt. The shackle and body shall withstand a tension pull of not less than 8,000 pounds when tested as specified in 4.4.1. The padlock shall withstand forceful opening for not less than 5 minutes when tested in accordance with 4.4.2. The operation of the padlock shall not be impaired, the shackle shall not release, and the padlock mechanism shall not be damaged when tested as specified in 4.4.8 and 4.4.9. The diameter of the padlock shackle shall be 7/16-inch ± 1/16-inch. The shackle clearance between the top of the case and the underside of the shackle of the type ES padlock shall be not more than 2 inches when the shackle is in the locked position. The shackle shall remain attached to the body of the padlock in the locked and in the unlocked conditions.

3.3.3 Padlock case. The padlock case shall be solid or laminated construction and its exterior shape may be in accordance with the manufacturer's commercial production. The assembled case, in the locked condition, shall have no openings other than the keyway and the shackle openings, except that drainage and assembly holes are permissible. When drainage holes are provided in the padlock, they shall be located in an area directly below or to the side of the shackle heel and toe and shall not exceed 0.093-inch diameter. Drainage holes shall not provide access to the tumblers, locking bolts, or plug retainers. The padlock case for the type ES padlock shall not exceed 9 cubic inches; and the case for the type SS padlock shall not exceed 12 cubic inches.

3.3.4 Padlock locking bolt. The locking bolt shall be a dead bolt type and shall engage the shackle at both heel and toe. The locking bolt may be one, two, or three piece construction and shall not depend upon spring action to hold the bolt in the locking position. End pressure on the locking bolt, when exerted by a shim or wedge type instrument, shall not move the bolt when tested as specified in 4.4.3.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T11:47Z
Check the source to verify that this is the current version before use.

3.3.5 Padlock locking mechanism. The padlock mechanism shall be pin tumbler, disc or wafer tumbler, side bar, or any other type or combination of types, provided the mechanism complies with all performance requirements of this specification. Padlock cylinders shall be removable and interchangeable within padlocks of the same manufacturer. Locks shall continue to meet requirements after an interchangeable cylinder has been inserted in the lock case. The cylinder shall be fastened within the padlock case by a method to prevent its forcible separation from the case and to prevent opening the padlock by the insertion of a shim or wedge type instrument along the side of the cylinder when tested as specified in 4.4.3. The method used to secure the locking mechanism within the case shall not be visible externally. It shall not be possible to remove the key from the padlock keyway when the cylinder is in the open position.

3.4 Padlock keys. Unless more than 2 keys are required (see 6.2), each padlock shall be furnished with two identical keys which shall operate the padlock. The keys shall be of a corrosion resisting metal having a minimum hardness of Rockwell B-75 when tested as specified in 4.4.7. The keys for individually keyed padlocks shall be identified by the same code designation. Individually keyed padlocks shall not be operable, except by the keys designed to operate the padlock, or be subject to key manipulation or key interchange when tested as specified in 4.4.5. The key shall turn easily and smoothly to operate the padlock without undue pressure or force and the key shall be tested for compliance with 4.4.6. Keys shall be delivered joined together by a nominal 1/2-inch diameter, key ring fabricated from not less than 0.032-inch steel wire.

3.5 Resistance to picking and bypassing techniques. Padlocks shall resist opening by picking and bypassing techniques for not less than the periods of time specified hereunder when tested in accordance with 4.4.3.

Technique	Minimum Elapsed Time
Picking	30 minutes
Bypassing	15 minutes

3.6 Resistant to heat. The padlock shall withstand the heat test in 4.4.4 without release of its shackle from the locked condition.

3.7 Lubrication. Padlocks shall be lubricated with either a powdered graphite conforming to SS-G-659, or with a powder conforming to MIL-M-7866.

3.8 Finish. The padlock's working surfaces shall have a smooth finish. All exterior and interior surfaces of the padlock shall be protected against corrosion. The case surface shall have a dull, smooth, chromium or cadmium finish, or for laminated construction, a tumbled finish of sufficient smoothness to accept marking required in 3.10. The quality of the finish and its application shall be in accordance with good commercial standards.

3.9 Chain and clevis for style I padlocks. When a chain with clevis is required, the chain shall conform to type I, grade C, class 5, style 1 of RR-C-271. The length of the chain shall be 9 inches, ± one link. The chain shall be securely fastened to the padlock case by a suitable clevis in a manner that will not interfere with the operation of the padlock. The attachment of the clevis to chain shall be by rivet or other equally secure fastening. The chain, clevis, and rivet shall be of corrosion resistant material.

3.10 Identification markings and numbers. The specification "FF-P-001480", the letters "US" and the name or trademark of the manufacturer shall be stamped, rolled or cast on the exterior surface of the padlock case. A random number shall be stamped in a distinctive, noncommercial code on the inside cylinder face. The lock and key shall bear serial numbers stamped on with a 3/32-inch die which will identify the lock cylinder and its applicable key. These numbers shall not disclose the key bitting either directly or by coding. All markings shall be neat, clear, and legible.

3.11 Workmanship. All parts shall be accurately fitted, well finished without sharp edges, burrs, and splinters. The finished padlock shall be a product able to withstand hard usage and shall be free from any defect which affects its appearance or operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:47Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. Except for qualification testing, the supplier is responsible for the performance of all inspection and test requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facility or service acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.1.2 Sampling for inspection. Except as otherwise indicated herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.1.3 End item inspection. A lot shall consist of all padlocks up to 5000 of the same type offered for inspection at one time. The sample unit for inspection shall be one completely assembled padlock.

4.1.4 Visual examination. The complete padlock shall be examined for defects classified in table I. The inspection level shall be S3 with an acceptable quality level (AQL) of 2.5 for major defects and 4.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects.

Examine	Defect	Classification	
		Major	Minor
Finish	Not as specified.		X
	Not uniformly finished or tumbled.		X
	Not smooth finish.		X
	Not free of burrs, splinters or sharp edges.		X
Design and construction	Part broken, split, fractured, or cracked.	X	
	Not as specified.	X	
	Any functioning component that is inoperative or will not operate as intended.	X	
	Not type specified.	X	
	Lock mechanism permits the shackle to be locked out in the open position.	X	
	Key can be removed when lock is in open position;	X	
	cylinder not removable or interchangeable.	X	
Case	Opening in case which will permit insertion of a device which will open lock.	X	
Chain, clevis and rivet (when specified)	Not attached securely.	X	
	Malformed, dented, buckled, bent out of shape.	X	
	Not length specified.	X	
	Not material specified	X	
Screws, nuts, and washers	Missing.	X	
Keys	Not as specified.	X	
	Amount of identical keys not as specified.	X	
	Identical keys not joined by a wire ring.		X
	Hardness not as specified.	X	
Identification marking	Missing, illegible, incorrect or incomplete.		X
	Not stamped, rolled, or cast as specified.		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:47Z
Check the source to verify that this is the current version before use.

4.1.5 Dimension and weight inspection. An inspection shall be made of the end item for compliance with dimensions and weights specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0, expressed in terms of defects per hundred units.

4.1.6 Testing for acceptance. Testing of the end items offered under contract or purchase order will be made by the Government from time to time to assure continued compliance with specification requirements. In this regard, samples from lots offered for acceptance will be taken by the Government inspector for testing as specified in 4.4.1 through 4.4.9. The testing shall be performed by a facility designated by the Government and at no cost to the supplier. Failure to meet testing requirments shall provide reason to suspend acceptance of the suppliers production until the Government inspector is satisfied that all defects in the product have been corrected.

4.2 Examination of Preparation for Delivery. An examination shall be made to determine that the packaging, packing and marking requirements of section 5 are complied with.

4.3 Qualification. In accordance with 3.1, products submitted for qualification shall be inspected for compliance with 4.I.4 and 4.1.5, and tested as specified in 4.4.1 through 4.4.9. The presence of any visual defect, any dimension not as specified, or failure to pass any test shall be cause for rejection of the QPL sample.

4.3.1. Testing costs. All costs entailed in testing a product for QPL purposes shall be borne by the manufacturer and shall be payable to the General Services Administration as directed by Standardization Division, Federal Supply Service.

4.3.2 Testing agency. Testing for QPL purposes shall be by an agency specifically designated by the General Services Administration.

4.3.3 Test sample. The QPL sample shall consist of 10 padlocks of the type the supplier proposes to furnish under this specification. The sample shall be forwarded, transportation prepaid, at a time and to a place designated by the Standardization Division, Federal Supply Service. Samples shall have tags attached which shall bear the specification symbol "FF-P-001480, padlock type, date of manufacture, the manufacturer's name and address, and the location of his plant. Tested samples will not be returned to the manufacturer.

4.3.4 Drawings and lists of materials. The manufacturer shall furnish four (4) complete sets of construction and assembly drawings and lists of materials for each type padlock submitted for qualification. These materials will be used during the qualification testing and for use afterward in inspections for acceptance of the product offered under a contract or purchase order. All material so furnished will be held in proprietary confidence. Once the product has been tested and approved for QPL, no changes of any kind shall be made in its construction or in the drawings unless prior written authorization to make changes is obtained from the General Services Administration.

4.3.5 Qualification tests. Qualification testing shall consist of the following tests which are described under Test methods in 4.4.

Test	Method	Requirement
Tension pull	4.4.1	3.3.2
Forceful opening	4.4.2	3.3.2
Picking and bypassing	4.4.3	3.3.4, 3.3.5, and 3.5
Key integrity	4.4.5	3.4
Key operation	4.4.6	3.4
Heat resistance	4.4.4	3.6
Material hardness	4.4.7	3.4
Jar test	4.4.8	3.3.2
Drop test	4.4.9	3.3.2

4.4 Test methods. For the purpose of the tests in 4.4.2 and 4.4.3, the following shall apply:

The padlock under test shall be locked by its shackle to a hasp or staple fastened to a solid upright member. Timing the test for record shall be accomplished by use of a stop watch or other suitable clocking device. The clocking device shall be started at the moment the test begins, and it shall not be stopped until the time limit applicable to the test has expired. In the event the padlock fails the test, the clock will be stopped at the point of failure.

The tests in 4.4.1 and 4.4.4 through 4.4.9 shall be performed as specified hereunder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:47Z
Check the source to verify that this is the current version before use.

4.4.1 Tension pull test. The body of the sample padlock under test shall be held in a device bearing against the top surface of the case without interfering with the shackle. A tension force of 8000 pounds shall be applied slowly along the vertical center line of the padlock shackle so as to put a direct and equal tension on each leg of the shackle to determine compliance with the tension pull requirements in 3.3.2. Pulling any shackle out of the locked position or the appearance of flaws or cracks in the shackle shall provide reason to consider that the padlock failed the test.

4.4.2 Forceful opening test. Forceful opening shall constitute any opening of the padlock within the time specified in 3.3.2 using any combination of the tools and equipment specified hereunder. The test shall be performed under the condition specified in 4.4, and the total weight of any combination of the allowable tools and equipment shall not exceed 10 pounds.

Bolt cutters, hand-powered, maximum 24-inch length.
Saws and drills, other than power driven.
Hammers, not to exceed 4-pound weight.
Torquing levers, not to exceed 18-inch length.
Heating equipment (limited to one tank of fuel).

4.4.3 Picking and bypassing test. The padlock shall be tested under the conditions specified in 4.4 and the bitting of the key for the padlock may be examined prior to the test. Picking and bypassing shall be limited to manual techniques. Opening the padlock by picking or bypassing techniques in less than the period of time specified in 3.5 shall provide reason to consider that the padlock failed the test.

4.4.4 Heat resistant test. The padlock shall be suspended by its shackle in the locked position in a heat treating oven. The oven temperature shall be regulated at not more than 1500°F. for 30 minutes. After 30 minutes the temperature shall be increased to 1800°F. for an additional 30 minutes. The temperature shall then be increased to not less than 2000°F. for an additional 60 minutes. After this 2 hour test cycle the specimen shall be cooled at normal room temperature and then examined for compliance with 3.6.

4.4.5 Key integrity test. A number of keys, other than keys designed and furnished by the manufacturer to operate the padlock, shall be inserted into the padlock under test. The padlock shall have failed the test if it can be opened by a key other than the key furnished by the manufacturer specifically designed for the padlock.

4.4.6 Key operation test. The padlock under test shall show no indication of wedging of the key in the keyway or excessive pressure to turn the key and unlock the shackle as specified in 3.4. The padlock shall be operated 10 times to make this determination.

4.4.7 Hardness test. Test for hardness shall be in accordance with ASTM E18. The average of three readings shall be within the limits specified in 3.4.

4.4.8 Jar test. The padlock shall be locked by its shackle to a staple fastened to a solid, wood upright member. The padlock case shall be held loosely with one hand and the padlock struck with a substantial blow, the hold on the padlock being released immediately before the blow, so as to jar the padlock forcibly against the wood upright. This procedure shall be performed at least 6 times, striking the padlock from different directions. A wood, plastic, or lead mallet weighing not more than 12 ounces shall be used to deliver the blows. After the test the padlock shall be examined for compliance with 3.3.2.

4.4.9 Drop test. The padlock shall be dropped six feet to a concrete surface or equal surface at least 10 times and then examined for compliance with 3.3.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each padlock with keys, and chain if applicable, shall be wrapped with material conforming to UU-P-553 and packaged in a close-fitting box conforming to PPP-B-566, style II, type A, class a, PPP-B-665, style B or PPP-B-676, type I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:47Z
Check the source to verify that this is the current version before use.

5.1.1.2 Intermediate packaging. Twelve unit packages of padlocks of like descriptions shall be intermediate packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636, class domestic. The fiberboard box shall be closed in accordance with appendix to the box specification.

5.1.2 Level B. The padlocks shall be packaged as specified in 5.1.1.1 and 5.1.1.2.

5.1.3 Level C. The padlocks shall be packaged to afford adequate protection against damage during shipment from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The padlocks in quantities as specified (see 6.2) shall be packed in a close-fitting box conforming to PPP-B-585, class 3; PPP-B-591; class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant or PPP-B-640, class 2, grade A. The wood boxes shall be provided with a case liner conforming to MIL-L-10547 and sealed in accordance with the appendix thereto. The boxes shall be closed and strapped in accordance with the specification or appendix thereto. The gross weight of the triple-wall fiberboard box and the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Six intermediate packages of padlocks of like descriptions shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The padlock shall be packed in insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplier. Containers shall be in accordance with the rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packaging and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the interior packaging and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The padlocks furnished under this specification are intended for use as determined by the using activity. The padlocks have a much higher degree of resistance to opening by force and covert techniques than do the average, key operated, general purpose padlocks.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.
(b) Type and style required (see 1.2.1).
(c) Number of keys when more than two are required (see 3.4).
(d) Levels of packaging, packing and marking required (see 5.1, 5.2, and 5.3).
(e) Quantity of intermediate packages required in the shipping container (see 5.2.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on the applicable Federal Qualified Products List, whether or not such products have actually been so listed by that date. The attention of suppliers is called to the requirement, and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification so that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406, and information pertaining to qualification may be obtained from that activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:47Z
Check the source to verify that this is the current version before use.